JOEL PEPPER

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EDUCATION

Drexel University *September 2019 – Present*
Ph.D. in Computer Science
M.S. in Computer Science *Awarded March 2022*
Advisor: Dr. David E. Breen

Member of Upsilon Pi Epsilon
Overall GPA: 3.99

The Ohio State University *August 2014 – August 2019*
B.S. in Biomedical Engineering
B.S. in Computer Science & Engineering

Member of Tau Beta Pi
Cum Laude (BME Degree)
Honors in Engineering (CSE Degree)
Overall GPA: 3.485

RESEARCH STATEMENT

Fields of interest: Computational Biology, Bioinformatics, Genomics, Biomedical Image Analysis

My thesis research is on the modeling and simulation of biological and bio-inspired structures. I have been developing a simulation framework for "4D materials," which includes materials that exhibit an active, time dependent warping response to some stimuli. This includes things like epithelial tissue, 4D prints (3D prints that warp when heated), virtual creatures and soft robots. My first publication on this topic detailed a novel 4D printing configuration involving multi-layered, cuboidal grids, and my second publication involved an extension of this framework to an epithelium-inspired grid of hexagonal prisms. Currently, I am working to extend the epithelium modeling capabilities to include cell-level, neural network-based agency, in order to enable cells to actively respond to their surroundings in a more biomimetic fashion.

Other areas of research I have been involved in outside of my dissertation project include cell and tissue engineering, image analysis, and genomics.

FUNDING

NSF Graduate Research Fellowship *September 2021 – September 2024*
Project proposal was entitled "Generalized Agent-Based Model of Responsive 4D Materials"

EMPLOYMENT HISTORY

Major Research Instrumentation: Accessible Data-Intensive Science *March 2025 – Present*
Doctoral Research Fellow *Philadelphia, PA*

- Funded researcher on an NSF sponsored project to development a platform for accessible data-intensive science and engineering

Institute for Data-Driven Dynamical Design *September 2024 – Present*
Doctoral Research Fellow *Philadelphia, PA*

- Funded researcher (under Dr. Jane Greenberg) on an NSF sponsored "Harnessing the Data Revolution" (HDR) institute working to automatically extract and study the contents of paper lab notebooks

Drexel Univ/Self *September 2021 – August 2024*
Doctoral Fellow *Philadelphia, PA*

· Externally funded via National Science Foundation Graduate Research Fellowship [1], [8]

Memorial Sloan Kettering Cancer Center *June 2023 – September 2023*
Computational Biology Intern *New York, NY*

· Research intern studying spatial epigenomics data under Dr. Christina Leslie

Drexel Univ Graduate College *September 2020 – December 2021*
Remote Course Facilitator *Philadelphia, PA*

· Assisted professors with the technical and logistical aspects of running online courses for four quarters during the pandemic

Biology Guided Neural Networks Project *September 2020 – September 2021*
Doctoral Research Fellow *Philadelphia, PA*

· Funded researcher (under Dr. Jane Greenberg) on an NSF HDR project; work centered around automatic generation of missing/deficient metadata for digital specimen images [3], [4], [6], [7], [9]

Drexel Univ Digital Development Camp *August 2020 – August 2020*
Counselor *Philadelphia, PA*

· Lead counselor for a web development camp sponsored by the college's Women in Tech Initiative

Drexel Univ Dept of Computer Science *September 2019 – August 2020*
Doctoral Teaching Fellow *Philadelphia, PA*

· Assisted in instruction and graded for graduate (CS foundations, advanced programming techniques) and undergraduate (freshman design, data structures) courses both in person and online

Drexel Univ REThink *July 2020 – July 2020*
Graduate Assistant *Philadelphia, PA*

· Assisted in the computer science department's NSF sponsored Research Experience for Teachers

Lab-Ally *November 2017 – September 2019*
Contract Software Developer *Columbus, OH*

· Developed software for an electronic lab notebook, focusing mainly on the server component

OSU Dept of Computer Science and Engineering *August 2016 – July 2019*
Student Instructional Assistant *Columbus, OH*

· Graded for and assisted in instruction of a web development precapstone course

OSU Biological and Applied Nanotechnology Lab *August 2015 May 2019*
Undergraduate Research Assistant *Columbus, OH*

· Worked on projects in nanotechnology and cell migration [10]

AWARDS

- (2023) College of Computing and Informatics Outstanding Graduate Student Award
- (2023) Phoebe Haas Doctoral Endowed Scholarship from the College of Computing and Informatics
- (2021) Best Student Paper Award for "Automatic Metadata Generation for Fish Specimen Image Collection" [9]

- (2020) Highly Commended/Honorable Mention for Drexel College of Computing and Informatics Graduate Teaching Assistant Excellence Award
- (2018) OSU College of Engineering Excellence in Capstone Design Award [5]

PUBLICATIONS

[1] J. Pepper and D. E. Breen, "An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets," in *Advances in Visual Computing*, ISSN: 1611-3349, Springer, Jan. 2025, pages 111–124, ISBN: 978-3-031-77392-1. DOI: 10.1007/978-3-031-77392-1_9.

[2] J. Pepper, E. Jones, X. Zhao, J. Furst, K. Langlois, et al., "AI-Ready Data: Knowledge Extraction from Archival Lab Notebooks," in *2024 IEEE International Conference on Big Data (BigData)*, Dec. 2024, pages 2489–2495. DOI: 10.1109/BigData62323.2024.10825206.

[3] M. A. Balk, J. Bradley, M. Maruf, . . ., J. Pepper, et al., "A FAIR and modular image-based workflow for knowledge discovery in the emerging field of imageomics," *Methods in Ecology and Evolution*, Apr. 2024, ISSN: 2041-210X. DOI: 10.1111/2041-210X.14327.

[4] D. E. Breen, A. Senin, A. Levere, J. Pepper, and J. Greenberg, "Specimen Outlining: A Computational Archival Science Approach," in *2023 IEEE International Conference on Big Data (BigData)*, Dec. 2023, pages 2004–2009. DOI: 10.1109/BigData59044.2023.10386985.

[6] J. Pepper, A. Senin, D. Jebbia, D. Breen, and J. Greenberg, "Metadata Verification: A Workflow for Computational Archival Science," in *2022 IEEE International Conference on Big Data (Big Data)*, Dec. 2022, pages 2565–2571. DOI: 10.1109/BigData55660.2022.10020340.

[7] K. Karnani, J. Pepper, Y. Baki, X. Wang, H. Bart Jr., D. E. Breen, and J. Greenberg, "Computational metadata generation methods for biological specimen image collections," en, *International Journal on Digital Libraries*, Nov. 2022, ISSN: 1432-1300. DOI: 10.1007/s00799-022-00342-1.

[8] J. Pepper and D. Breen, "Design and Simulation of Multi-Tiered 4D Printed Objects," *Computer-Aided Design*, volume 20, number 3, pages 489–506, Sep. 2022. DOI: 10.14733/cadaps.2023.489-506.

[9] J. Pepper, J. Greenberg, Y. Baki, X. Wang, H. Bart, and D. Breen, "Automatic Metadata Generation for Fish Specimen Image Collections," in *2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, Sep. 2021, pages 31–40. DOI: 10.1109/JCDL52503.2021.00015.

[10] Y. Cui, S. Cole, J. Pepper, J. J. Otero, and J. O. Winter, "Hyaluronic acid induces ROCK-dependent amoeboid migration in glioblastoma cells," en, *Biomaterials Science*, volume 8, number 17, pages 4821–4831, Aug. 2020, ISSN: 2047-4849. DOI: 10.1039/D0BM00505C.

PATENT

[5] Z. Ali, P. Scott, A. Kiourti, B. Zhang, J. Pepper, et al., "System and method for completing a measurement of anxiety," Granted Patent US 11571155 B2, Feb. 7, 2023. [Online]. Available: `https://lens.org/025-853-906-337-42X`.

SOURCE CODE

- Source code for [1]: To Be Released
- Source code for [6], [7], [9]: hdr-bgnn/drexel_metadata
- Source code for [8]: DrJPepper/4D-Multilayer-Modeler

PRESENTATIONS

- Workshop presentation, "Automatic Metadata Generation and Verification for AI-Ready Fish Image Collections," Imageomics Workshop @ The Association for the Advancement of AI Conference, Philadelphia, PA, March 2025 (Poster)

- Workshop paper presentation, "AI-Ready Data: Knowledge Extraction from Archival Lab Notebooks", IEEE Big Data Computational Archival Science Workshop, Washington, DC, December 2024 [2] (Original Video Link, Mirror, Slides)
- Paper presentation, "An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets", International Symposium on Visual Computing, Stateline, NV, October 2024 [1] (Recorded Practice Run)
- Poster presentation given by REU student, "AI-Ready Data: Knowledge Extraction from Laboratory Notebooks," Northwestern University ID4 Symposium, Evanston, IL, August 2024 (Poster)
- Poster presentation, "An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets," Drexel Emerging Graduate Scholars Conference, Philadelphia, PA, April 2024 (Poster)
- Workshop presentation, "Enumerating and Addressing AI-Readiness Challenges from the Biology Guided Neural Networks HDR Project," AI-Ready Data: Navigating the Dynamic Frontier of Metadata and Ontologies, Philadelphia, PA, April 2024 (Slides)
- Hackathon participant, "Evaluating the effects of Hi-C and TF on cis-regulation of gene expression," 4D Nucleome Hackathon, Seattle, WA, March 2024 (Presentation Slides)
- Paper presentation, "Metadata Verification: A Workflow for Computational Archival Science" (with Andrew Senin), IEEE Big Data Conference, Online, December 2022 [6] (Video)
- Paper presentation, "Design and Simulation of Multi-Tiered 4D Printed Objects," CAD Conference, Online, July 2022 [8] (Recorded Practice Run)
- Workshop, "3D Data Visualization with VTK," Philly Codefest (run by Drexel University), Online, March 2022 (Original Video Link, Mirror, Demo Code)
- Paper presentation, "Automatic Metadata Generation for Fish Specimen Image Collections," ACM / IEEE Joint Conference on Digital Libraries, Online, September 2021 [9] (Video)
- Presenter and panelist, Taxonomic Names and Metadata: A Framework for Big Data Interoperability Session, Great Lakes Bioinformatics Conference, Online, May 2021
- Talk, "Approaches for Computing Specimen Image Research Data" (with Jane Greenberg and David Breen), International Conference on Statistical Tools and Techniques for Research Data Analysis, Online, January 2021

MENTORING

- Elizabeth Jones, Summer REU, June 2024 – August 2024 (Poster)
- Andrew Senin, Undergraduate Researcher and Senior Design Student, August 2022 – December 2023 [4], [6]
- Kevin Karnani, Undergraduate Research Co-Op, September 2021 – March 2022 [7]

TECHNICAL STRENGTHS

Environment	GNU/Linux user of 10+ years with extensive knowledge of its system administration, networking and scripting
Computer Languages	C++, C, Java, Python, HTML/CSS, JavaScript, SQL, Bash, MATLAB
Tools | Libraries	Git, Vim, Make, CMake | PyTorch, OpenCL, Arduino